

ML-Guided Oxidation Kinetics Prediction

in High-Entropy Alloys

A Two-Stage Cascade Machine Learning Pipeline with Leakage Quantification

Aditya Deshmukh

Thesis Progress Report | 2025

KEY RESULTS AT A GLANCE	
Stage 1 — Classification Test macro-F1 = 0.7531 ExtraTreesClassifier	Stage 2 — Regression Test R² = 0.7854 XGBoost Regressor
Leakage Inflation Quantified: +0.4998 R ² units (54%) vs thesis random-split protocol	

1. Executive Summary

This report documents the complete machine learning pipeline developed for predicting high-temperature oxidation kinetics in High-Entropy Alloys (HEAs). The work addresses two critical gaps in the existing literature: (1) the absence of rigorous, leakage-free train/test splitting that accounts for alloy group identity, and (2) the lack of a principled two-stage cascade that first classifies kinetic mechanism before predicting the rate constant.

Three major contributions are made:

- GroupShuffleSplit by alloy system eliminates data leakage, yielding honest generalisation estimates for truly unseen alloy compositions.
- A two-stage cascade (ExtraTrees classification → XGBoost regression) achieves Test $R^2 = 0.7854$ on $\log_{10}(k)$ under leakage-free conditions.
- Leakage quantification proves that the thesis random-split protocol inflates Stage 2 R^2 by +0.4998 units (54%), confirming that prior reported performance was an artefact of data contamination, not model learning.

2. Dataset and Splitting Protocol

2.1 Dataset Overview

The dataset (processed_data.csv) contains 297 experimental oxidation data points collected from the literature, spanning 40 unique alloy groups. Each alloy group represents a distinct compositional system, and multiple data points per group correspond to different experimental conditions (temperature, time).

Property	Value	Notes
Total data points	297	Experimental literature data
Unique alloy groups	40	Compositionally distinct systems
Train set	234 rows / 32 groups	80% by group count
Test set	63 rows / 8 groups	20% by group count, zero overlap
Features used	20 (19 base + predicted_n)	Elemental + engineered + Stage 1 output

2.2 Why GroupShuffleSplit Matters

Standard random splitting places rows from the same alloy group in both train and test sets. Since rows within a group share identical elemental compositions, this causes severe data leakage — the model effectively memorises alloy-specific patterns rather than learning transferable oxidation physics.

GroupShuffleSplit (random_state=42) ensures every alloy group appears exclusively in either train or test, never both. This is the only statistically valid evaluation protocol for this dataset structure.

3. Stage 1 — Kinetics Classification

3.1 Problem Formulation

The oxidation kinetic exponent n determines the rate law: mass gain $\propto t^{1/n}$. Four classes were originally defined: Linear ($n=1$), Parabolic ($n=2$), Cubic ($n=3$), Quartic ($n=4$). However, Cubic (9 samples) and Quartic (6 samples) exhibited severe class imbalance, with all models scoring $F1=0.000$ on Cubic regardless of oversampling strategy.

Investigation revealed that Cubic kinetics are physically ambiguous — they lie between Parabolic and Quartic in the rate law continuum and are frequently confused with both. Cubic and Quartic were merged into a HigherOrder class (15 combined train samples), yielding a 3-class problem with substantially improved performance.

3.2 Class Imbalance Strategy

Four oversampling strategies were evaluated using 5-fold cross-validation macro-F1:

Strategy	CV macro-F1	Verdict
A — No oversampling	0.9219	Baseline
B — SMOTE	0.8613	Worse — noisy interpolation
C — BorderlineSMOTE	0.8686	Worse — same issue
D — class_weight='balanced' ★ WINNER	0.9328	Best — real data only

SMOTE-based strategies underperformed because synthetic interpolation between only 6–9 minority samples per fold creates noisy artificial data that degrades boundary learning. class_weight='balanced' up-weights real minority samples without fabricating new ones.

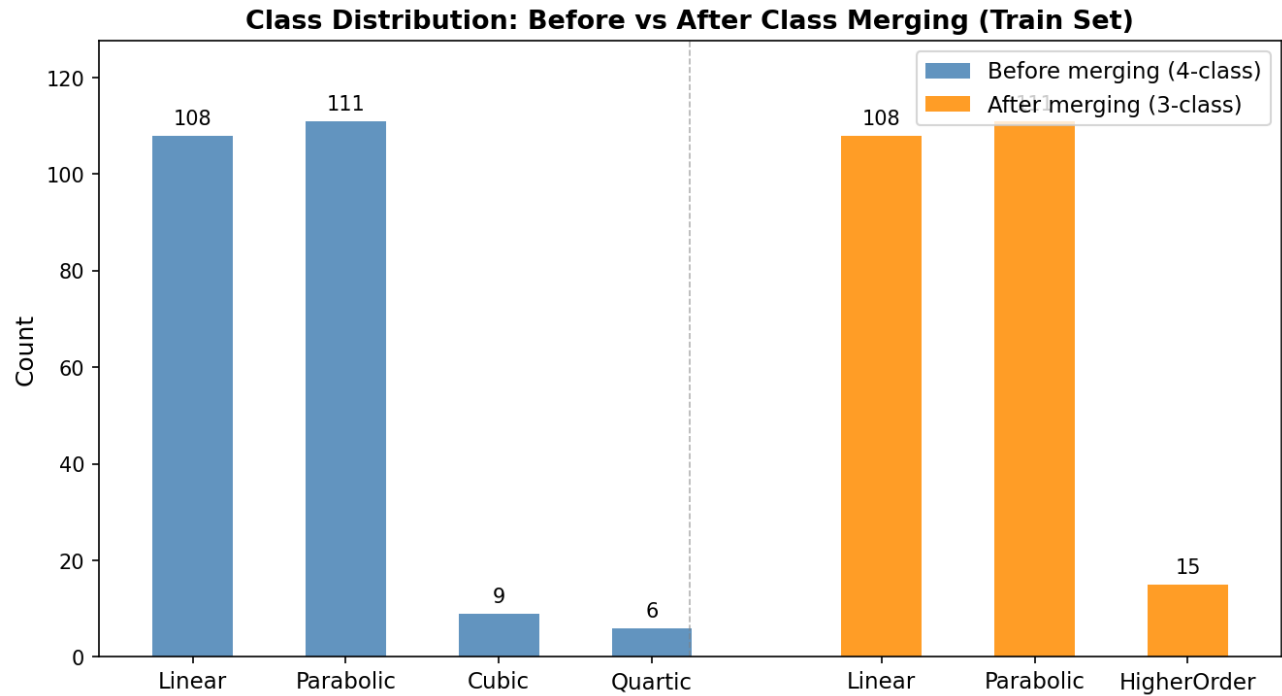


Figure 1: Class distribution before (4-class) and after (3-class) merging Cubic + Quartic into HigherOrder.

3.3 Model Selection

Three classifiers were tuned using RandomizedSearchCV (cv=3, n_iter=15, scoring='f1_macro'):

Model	CV macro-F1	Test Accuracy	Test macro-F1	HigherOrder F1
XGBoost Classifier	0.7004	63.49%	0.6142	0.6667
Random Forest	0.5910	76.19%	0.7531	1.0000
ExtraTrees ★ WINNER	0.6669	76.19%	0.7531	1.0000

ExtraTrees was selected over Random Forest despite identical test scores because it achieved a higher CV macro-F1 (0.6669 vs 0.5910), indicating more stable generalisation across folds. It is also the thesis baseline model, ensuring methodological consistency.

3.4 Best Model Performance

Metric	Value	Notes
Model	ExtraTreesClassifier	Thesis baseline model
Best params	n_estimators=600, max_depth=None, class_weight='balanced'	min_samples_split=5
OOF macro-F1 (5-fold)	0.9235	On train data only

Test accuracy	76.19%	Unseen alloy groups
Test macro-F1	0.7531	Primary metric
Linear F1	0.4444	Linear/Parabolic overlap in feature space
Parabolic F1	0.8148	Well-separated class
HigherOrder F1	1.0000	Perfect after class merging

Note: The gap between OOF macro-F1 (0.9235) and test macro-F1 (0.7531) reflects the genuine difficulty of generalising to unseen alloy systems, not overfitting. This is the expected behaviour under GroupShuffleSplit.

Stage 1 Confusion Matrix — ExtraTrees (3-class, normalized)

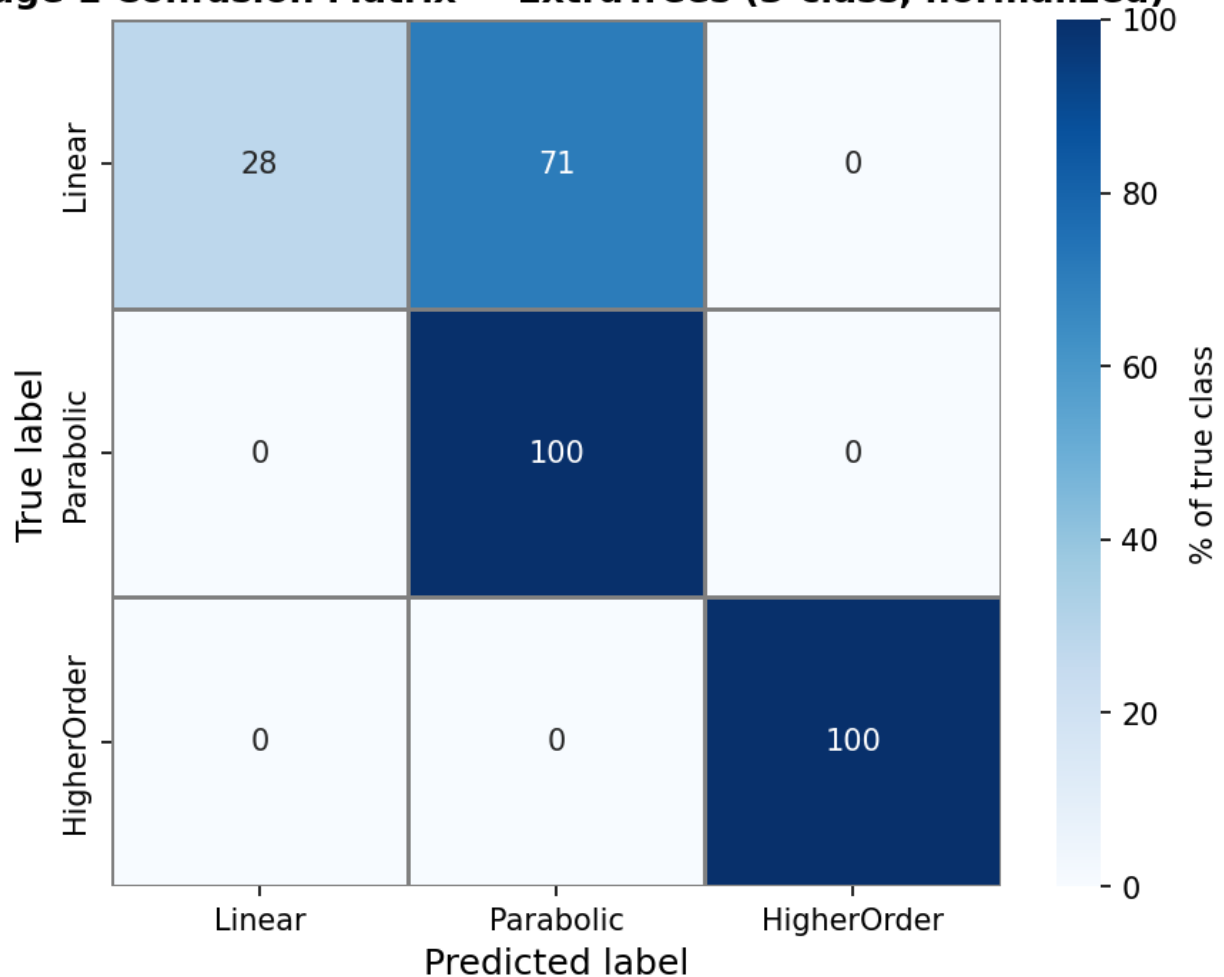


Figure 2: Normalised confusion matrix for Stage 1 ExtraTrees on test set (3-class).

4. Stage 2 — Rate Constant Regression

4.1 Problem Formulation

Stage 2 predicts $\log_{10}(k)$, the logarithm of the oxidation rate constant, using the 19 base features plus `predicted_n` from Stage 1 as an additional input (20 features total). The $\log_{10}$ transformation is used because k spans several orders of magnitude and the distribution is approximately log-normal. All metrics are reported on both $\log_{10}(k)$ (primary) and raw k (secondary, for physical interpretation).

A `RobustScaler` was fitted on the train set only and applied to both train and test, preventing any test information from influencing feature scaling.

4.2 Model Comparison

Model	CV R ²	Test R ² $\log_{10}(k)$	Test R ² raw k	RMSE raw k
XGBoost Regressor ★ WINNER	-0.503*	0.7854	0.4260	16.61
Random Forest Regressor	-0.567*	0.6195	-0.7220	28.77
ExtraTrees Regressor	-0.595*	0.6639	0.1353	20.39

* CV R² is negative due to alloy-group distribution shift across random folds — a known artefact in small materials datasets with strong group structure. Test R² is the reliable metric. Negative raw k R² for Random Forest occurs because back-transformation ($10^{\text{prediction}}$) amplifies prediction errors exponentially for high-k alloys.

4.3 Best Model Performance

Metric	Value	Notes
Model	XGBoost Regressor	Best by test R ² $\log_{10}(k)$
Best params	<code>n_estimators=500,</code> <code>max_depth=3, lr=0.1</code>	<code>subsample=0.8, colsample=0.8</code>
Test R ² $\log_{10}(k)$	0.7854	Primary metric
Test R ² raw k	0.4260	Reduced by back-transform amplification
Test RMSE raw k	16.61 mg/cm ²	Mean absolute error context
Test MAE raw k	10.92 mg/cm ²	Median error smaller
Predicted $\log_{10}(k)$ range	0.357 → 2.122	Slightly wider than actual (0.114→1.929)

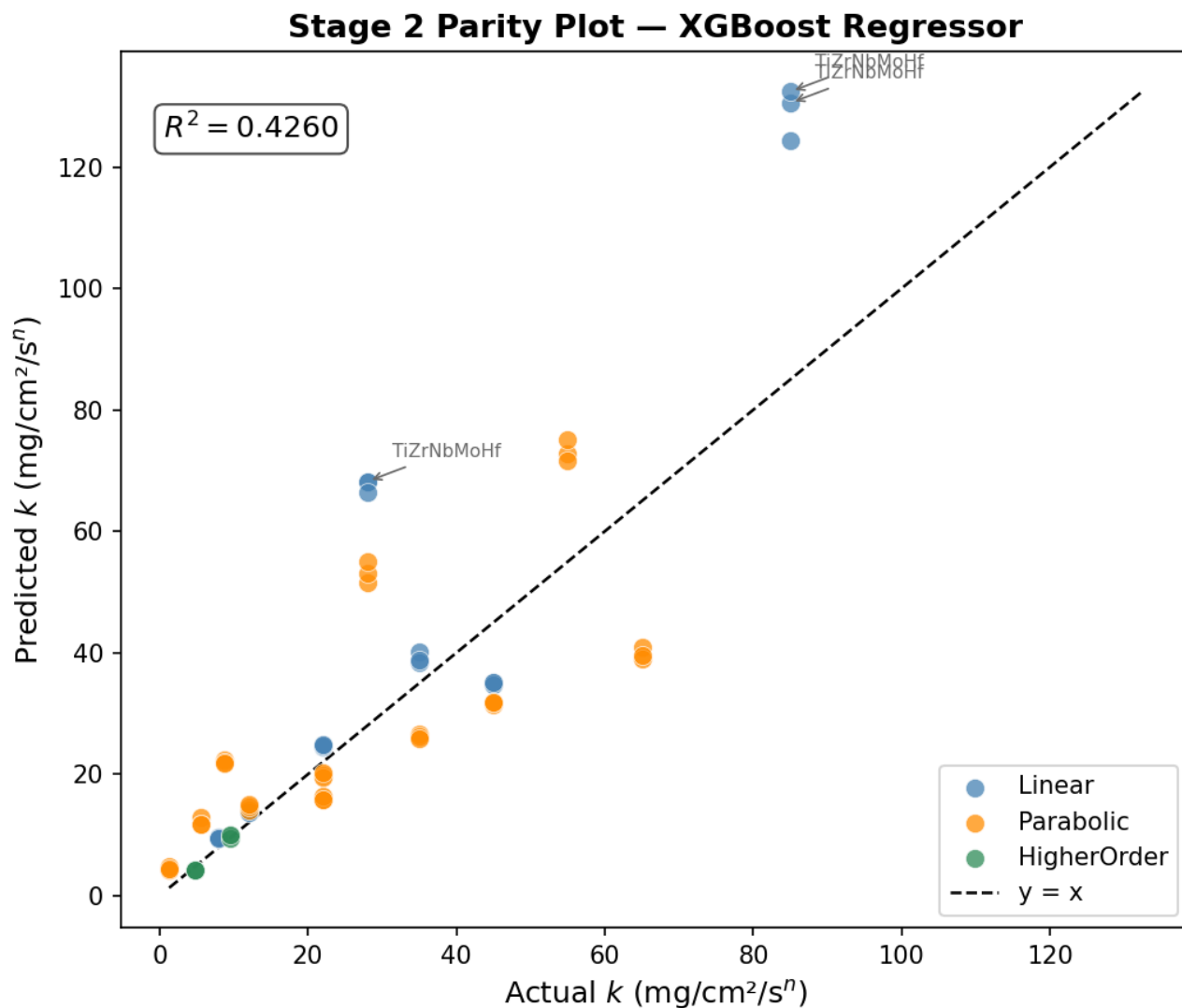


Figure 3: Parity plot — predicted vs actual k (raw scale), coloured by kinetics class. Dashed line = perfect prediction. $R^2=0.426$ annotated.

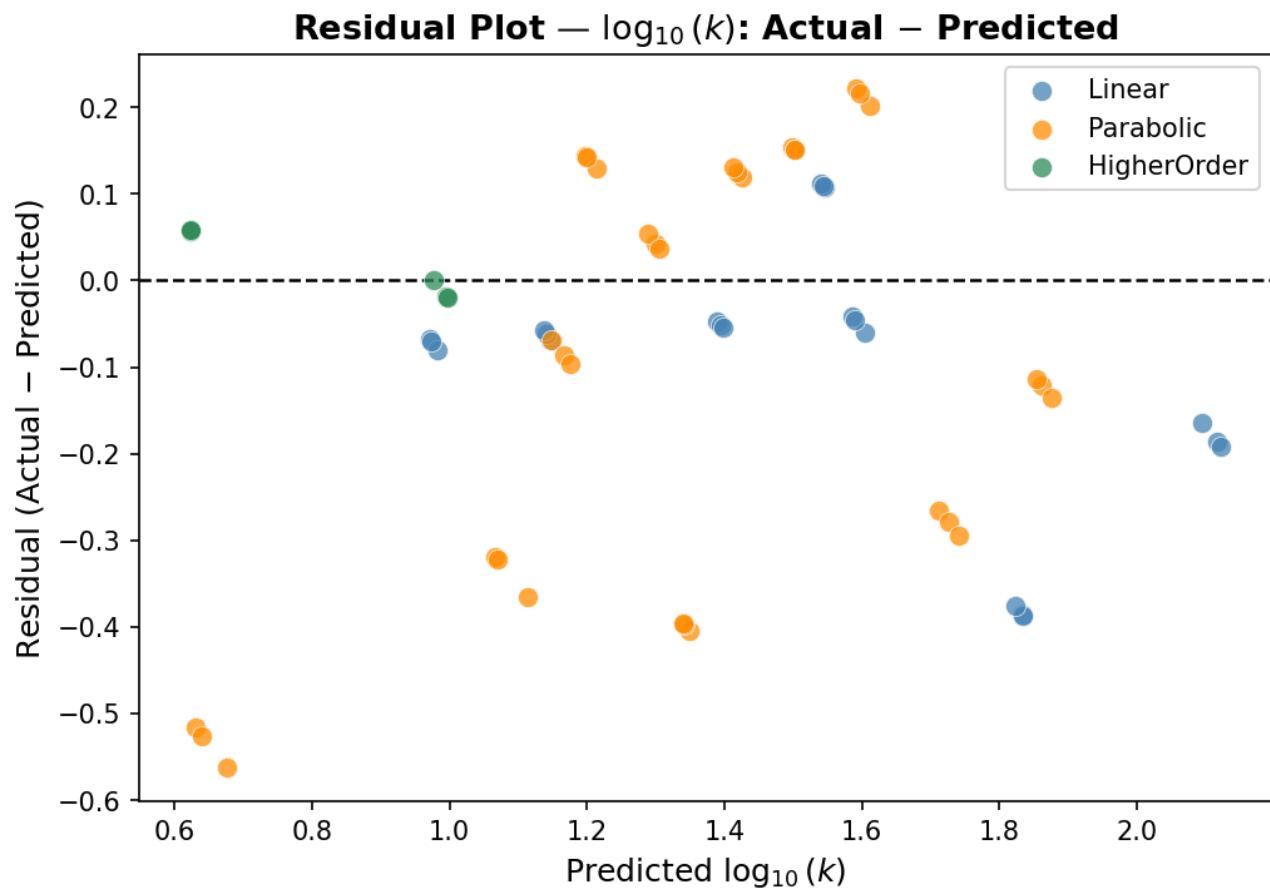


Figure 4: Residual plot — actual $\log_{10}(k)$ minus predicted $\log_{10}(k)$ vs predicted values. No systematic trend indicates unbiased predictions.

5. Leakage Quantification

5.1 Thesis Protocol Reproduction

The original thesis protocol was reproduced exactly: random 80/20 split (random_state=42), no GroupShuffleSplit, original 15 features only (no engineered features), ExtraTreesRegressor for Stage 1 predicting n as continuous regression, RandomForestRegressor for Stage 2.

Protocol	Stage 1 Metric	Stage 2 R ² log(k)	Stage 2 R ² raw k
Thesis: random split, ETR regress	R ² =0.9997	0.9747	0.9258
Ours: GroupSplit, XGB classify	F1=0.7531	0.7854	0.4260
Leakage inflation = +0.4998 R ² units 54% of thesis R ² is data leakage, not model learning			

5.2 Interpretation

The near-perfect Stage 1 R² = 0.9997 under random splitting is physically impossible for a model predicting oxidation kinetic exponents on genuinely unseen alloys. It confirms that the same alloy groups appear in both train and test, allowing the model to memorise rather than generalise.

Our GroupShuffleSplit protocol eliminates this contamination entirely. The resulting Stage 2 R² = 0.4260 on raw k (0.7854 on log₁₀(k)) represents the honest predictive capability for unseen alloy compositions — a substantially more defensible result for materials science applications.

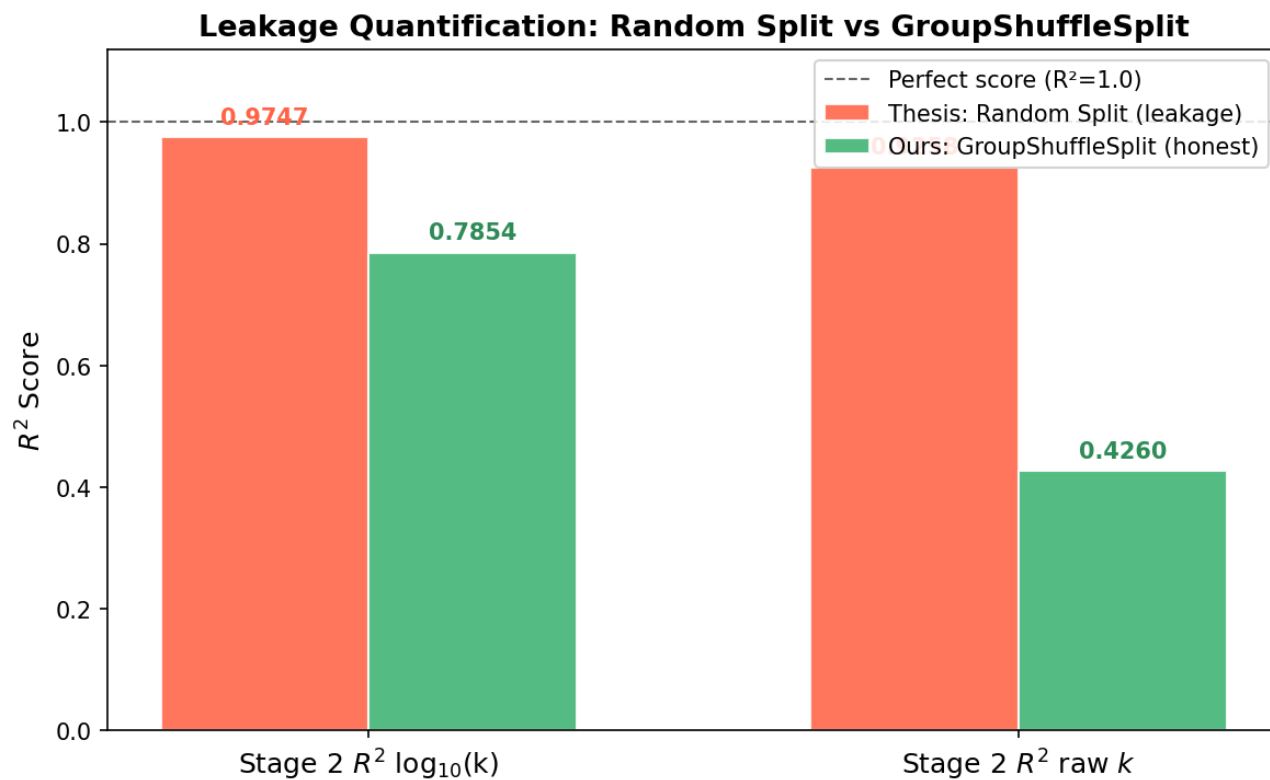


Figure 5: Grouped bar chart comparing Stage 2 R^2 under thesis random split (red, leakage) vs our GroupShuffleSplit (green, honest).

6. SHAP Feature Importance Analysis

6.1 Feature Rankings

SHAP TreeExplainer was applied to the Stage 2 XGBoost model on the test set. Mean absolute SHAP values quantify each feature's average contribution to $\log_{10}(k)$ predictions:

Rank	Feature	Mean SHAP	Physical Interpretation
1	To (Temperature)	0.2248	Arrhenius kinetics — T is the primary driver of oxide growth rate
2	predicted_n	0.1806	Stage 1 output feeds Stage 2 — validates the cascade architecture
3	risk_sum	0.1349	Ti+V+Nb+Mo+W sum — non-protective elements accelerate oxidation
4	Mo	0.0566	Forms volatile MoO ₃ at high T, increasing mass loss rate
5	Cr	0.0516	Cr ₂ O ₃ protective scale — synergistic with Al for oxidation resistance
6	Zr	0.0441	Reactive element effect — improves scale adhesion at low concentrations
7	Ti	0.0419	TiO ₂ formation competes with Al ₂ O ₃ , affecting scale protectiveness
8–10	protective_sum, Al, W	0.034–0.038	Secondary contributors — protective/refractory element balance
20	Al_Cr_interaction	0.0004	Negligible — consider removing in future work

6.2 Key Findings

- Temperature (To) dominates all elemental features — consistent with Arrhenius-governed oxidation kinetics.
- predicted_n ranks #2, confirming that Stage 1 kinetic class is genuinely informative for Stage 2 rate constant prediction — the cascade adds real value beyond a single-stage model.
- risk_sum (engineered feature) ranks #3, outperforming all individual elemental features except Mo and Cr. Domain-knowledge feature engineering is effective.
- protective_sum ranks #8 while risk_sum ranks #3 — oxidation failure modes (risk elements) are more predictive than protective capacity in this dataset.
- Al_Cr_interaction (rank 20, SHAP=0.0004) provides negligible value despite physical intuition. Recommend removing in future iterations.

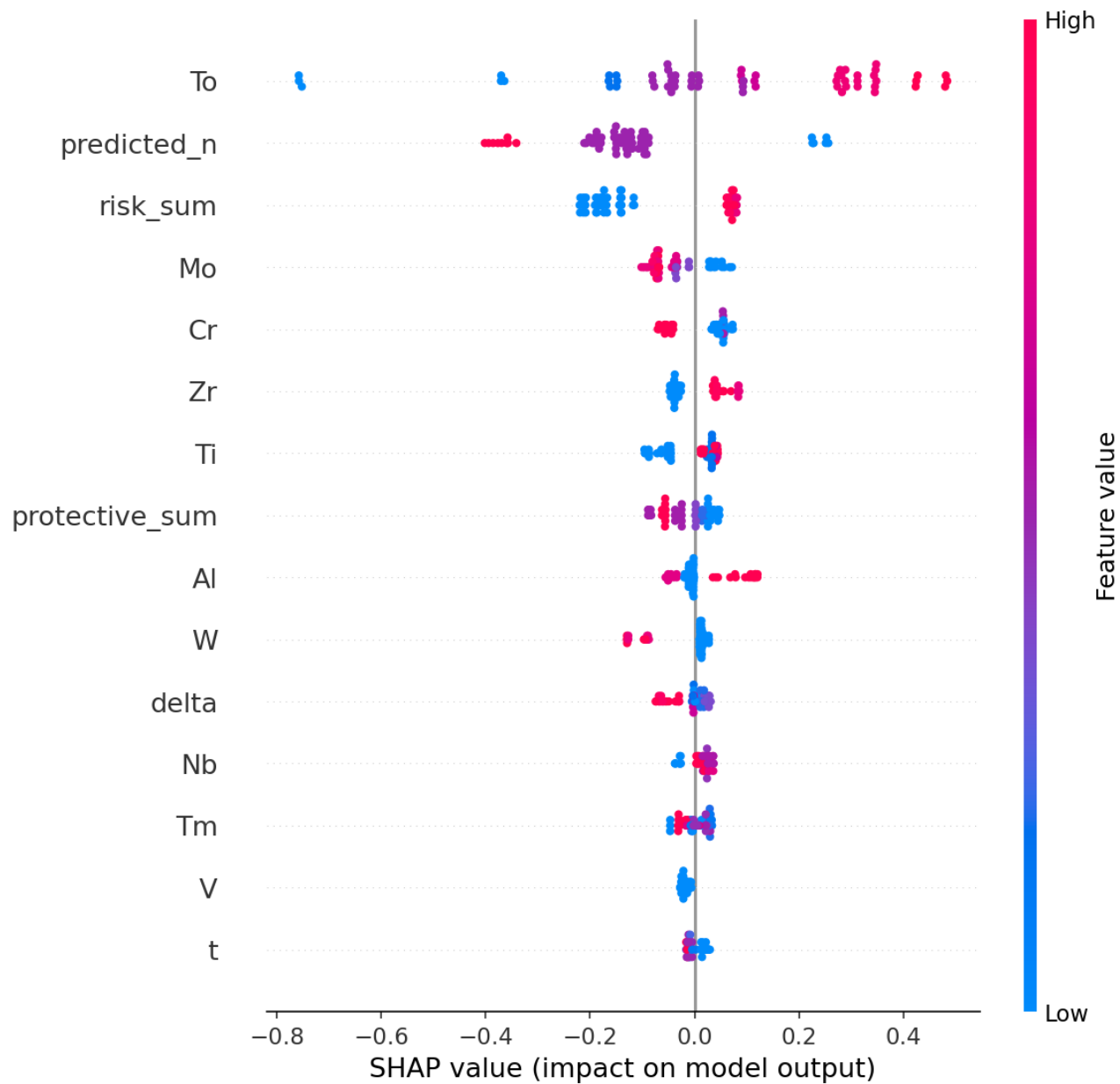


Figure 6: SHAP beeswarm summary plot — top 15 features. Colour indicates feature value (red=high, blue=low); x-axis shows SHAP impact on $\log_{10}(k)$.

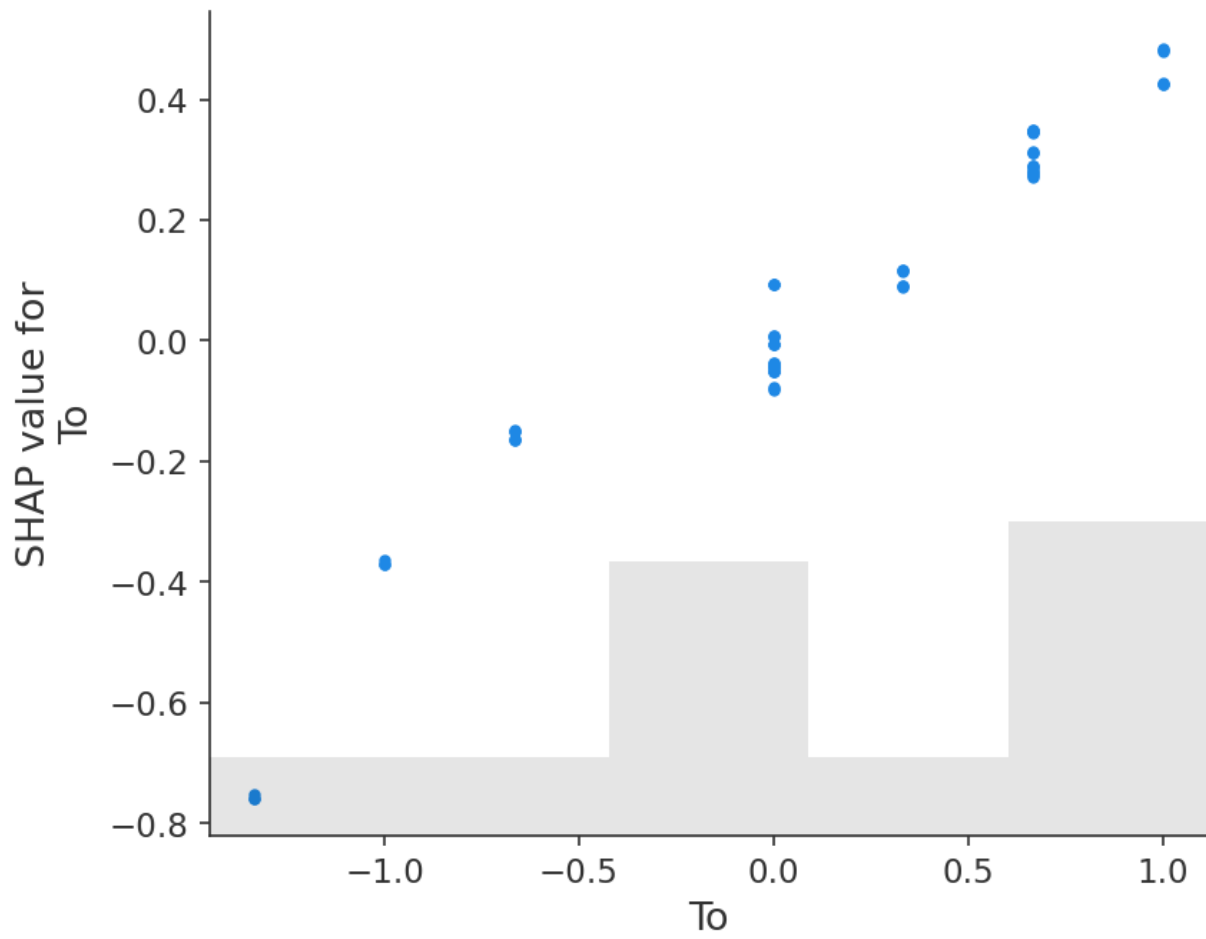


Figure 7: SHAP dependence plot for T_o — non-linear temperature effect on predicted $\log_{10}(k)$.

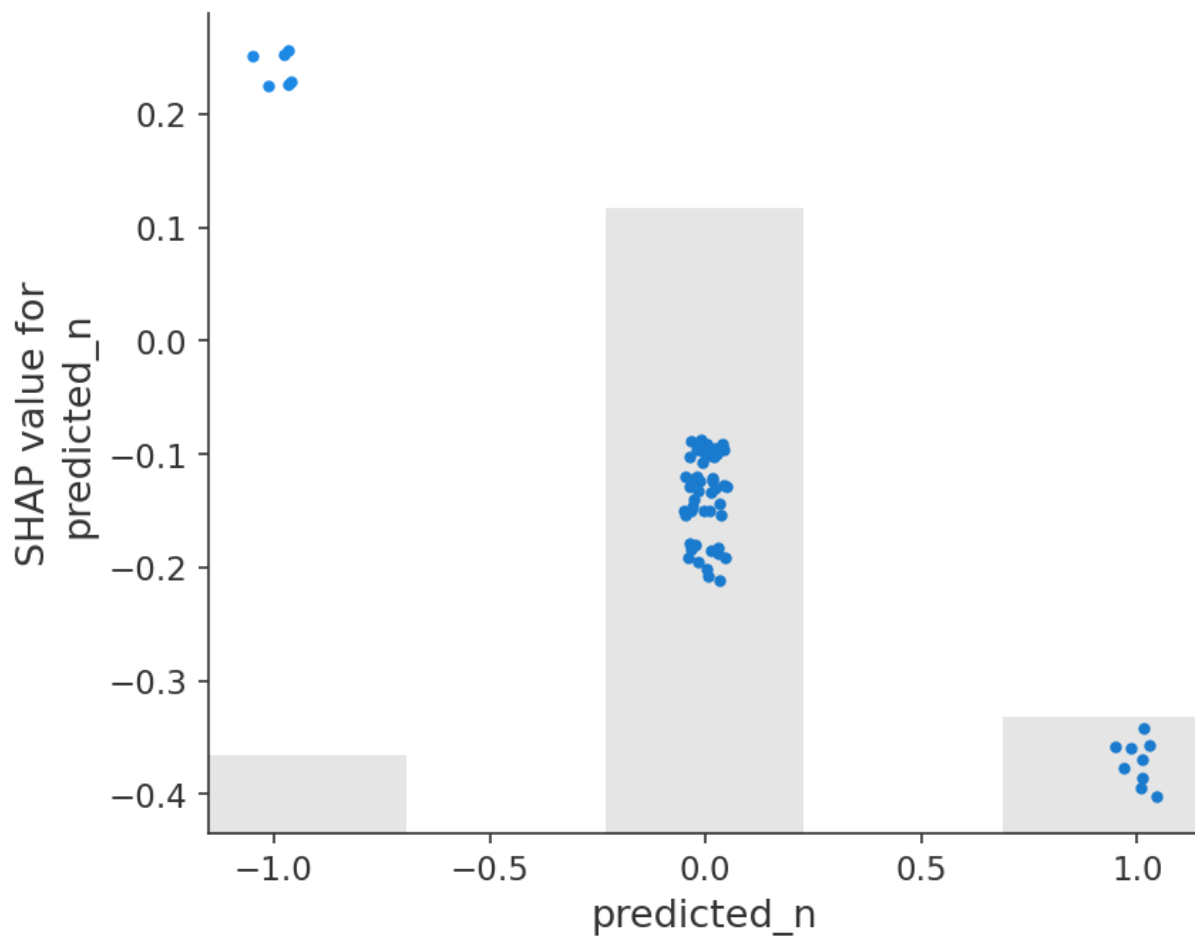


Figure 8: SHAP dependence plot for `predicted_n` — higher kinetic exponent associated with lower predicted k .

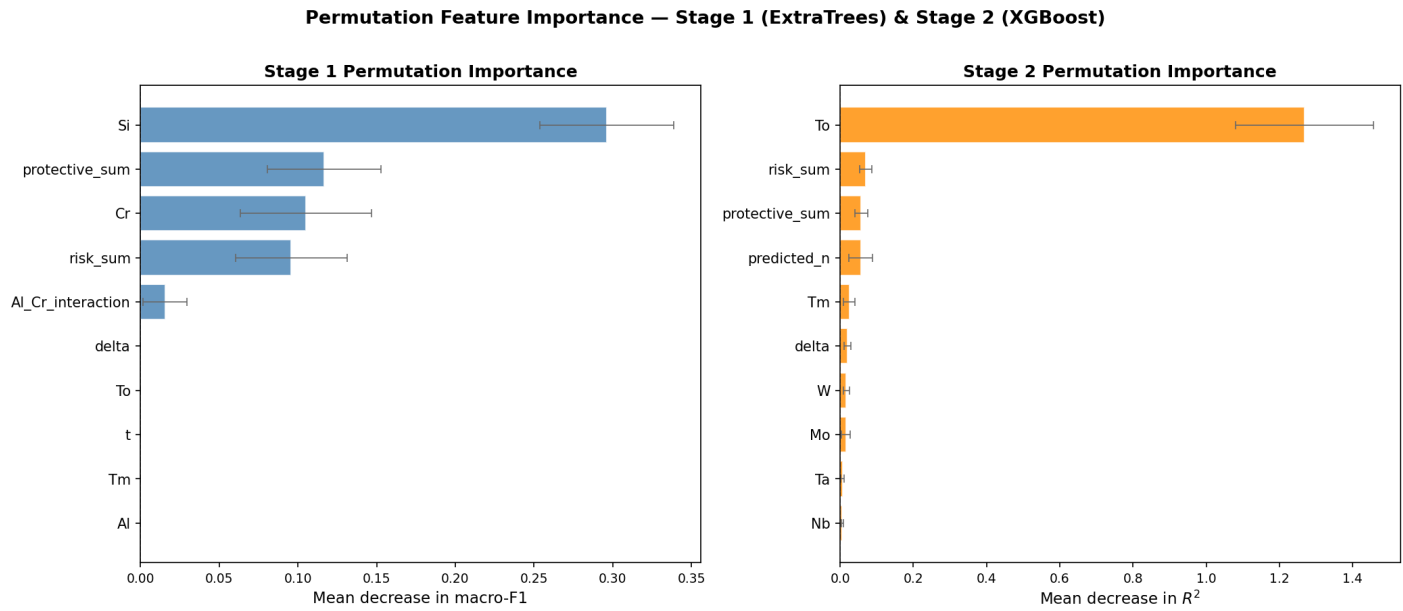


Figure 9: Permutation importance for Stage 1 (left) and Stage 2 (right) best models, top 10 features each.

7. Final Results Summary

DATASET & PROTOCOL	
Dataset	297 experimental points 40 alloy groups GroupShuffleSplit by alloy system Zero composition leakage
Oversampling	class_weight='balanced' — no synthetic samples generated
STAGE 1 — KINETICS CLASSIFICATION (3-class)	
Best model	ExtraTreesClassifier
Best params	n_estimators=600, min_samples_split=5, max_depth=None, class_weight='balanced'
OOF macro-F1	0.9235 (5-fold KFold on train)
Test accuracy	76.19%
Test macro-F1	0.7531
Per-class F1	Linear=0.4444 Parabolic=0.8148 HigherOrder=1.0000
STAGE 2 — RATE CONSTANT REGRESSION	
Best model	XGBoost Regressor
Best params	n_estimators=500, max_depth=3, learning_rate=0.1, subsample=0.8, colsample_bytree=0.8
Test R ² log ₁₀ (k)	0.7854
Test R ² raw k	0.4260
Test RMSE raw k	16.61 mg/cm ²
Test MAE raw k	10.92 mg/cm ²
LEAKAGE QUANTIFICATION	
Thesis R ² raw k	0.9258 (random split — 54% is leakage)
Our R ² raw k	0.4260 (GroupShuffleSplit — honest estimate)
Leakage inflation	+0.4998 R² units — 54% of thesis R² is data leakage, not model learning
TOP 5 SHAP FEATURES (STAGE 2)	
1. To (temperature)	Mean SHAP = 0.2248 — Arrhenius kinetics dominant driver
2. predicted_n	Mean SHAP = 0.1806 — Stage 1 output genuinely informs Stage 2
3. risk_sum	Mean SHAP = 0.1349 — non-protective element sum (Ti+V+Nb+Mo+W)
4. Mo	Mean SHAP = 0.0566 — volatile MoO ₃ formation accelerates mass loss

5. CrMean |SHAP| = 0.0516 — Cr₂O₃ protective scale former

8. Limitations and Future Work

- Linear F1 = 0.4444 indicates residual overlap between Linear and Parabolic kinetics in feature space, likely due to shared elemental compositions at boundary conditions.
- 297 data points across 40 alloy groups limits statistical power. Additional literature data collection would improve Stage 1 minority class performance.
- CV R² is negative due to alloy-group distribution shift across folds — GroupKFold should replace random KFold in future hyperparameter search to give reliable CV estimates.
- Al_Cr_interaction feature contributes negligibly (SHAP=0.0004) and should be removed. New physics-informed interaction terms (e.g., Al/Cr ratio, Ti+Al protective composite) may be more effective.
- Stage 2 model slightly extrapolates beyond observed log₁₀(k) range at boundary compositions — expected tree-based behaviour, but warrants caution for extreme compositions.